

Abramowitz and Stegun (AS) is the informal name of a 1964 mathematical reference work edited by Milton Abramowitz and Irene Stegun of the United States National Bureau of Standards (NBS), now the National Institute of Standards and Technology (NIST). Its full title is Handbook of Mathematical Functions with Formulas, Graphs, and Mathematical Tables. A digital successor to the Handbook was released as the "Digital Library of Mathematical Functions" (DLMF) on 11 May 2010, along with a printed version, the NIST Handbook of Mathematical Functions, published by Cambridge University Press.

Overview

Since it was first published in 1964, the 1046-page Handbook has been one of the most comprehensive sources of information on special functions, containing definitions, identities, approximations, plots, and tables of values of numerous functions used in virtually all fields of applied mathematics. The notation used in the Handbook is the de facto standard for much of applied mathematics today.

At the time of its publication, the Handbook was an essential resource for practitioners. Nowadays, scientific calculators, numerical analysis software packages, and computer algebra systems have replaced the function tables, but the Handbook remains an important reference source. The foreword discusses a meeting in 1954 in which it was agreed that "the advent of high-speed computing equipment changed the task of table making but definitely did not remove the need for tables".

More than 1,000 pages long, the Handbook of Mathematical Functions was first published in 1964 and reprinted many times, with yet another reprint in 1999. Its influence on science and engineering is evidenced by its popularity. In fact, when New Scientist magazine recently asked some of the world's leading scientists what single book they would want if stranded on a desert island, one distinguished British physicist said he would take the Handbook.

The Handbook is likely the most widely distributed and most cited NIST technical publication of all time. Government sales exceed 150,000 copies, and an estimated three times as many have been reprinted and sold by commercial publishers since 1965. During the mid-1990s, the book was cited every 1.5 hours of each working day. And its influence will persist as it is currently being updated in digital format by NIST.

The chapters are:

Mathematical Constants

Physical Constants and Conversion Factors

Elementary Analytical Methods

Elementary Transcendental Functions

Exponential Integral and Related Functions

Gamma Function and Related Functions

Error Function and Fresnel Integrals

Legendre Functions

Bessel Functions of Integral Order

Bessel Functions of Fractional Order

Integrals of Bessel Functions

Struve Functions and Related Functions

Confluent Hypergeometric Functions

Coulomb Wave Functions

Hypergeometric Functions

Jacobian Elliptic Functions and Theta Functions

Elliptic Integrals

Weierstrass Elliptic and Related Functions

Parabolic Cylinder Functions

Mathieu Functions

Spheroidal Wave Functions

Orthogonal Polynomials

Bernoulli and Euler Polynomials, Riemann Zeta Function

Combinatorial Analysis

Numerical Interpolation, Differentiation, and Integration

Probability Functions

Miscellaneous Functions

Scales of Notation

Laplace Transforms

Editions

Because the Handbook is the work of U.S. federal government employees acting in their official capacity, it is not protected by copyright in the United States. While it could be ordered from the Government Printing Office, it has also been reprinted by commercial publishers, most notably Dover Publications (ISBN 0-486-61272-4), and can be legally viewed on and downloaded from the web.

While there was only one edition of the work, it went through many print runs including a growing number of corrections.

Original NBS edition:

1st printing: June 1964; errata:

2nd printing with corrections: November 1964; errata:

3rd printing with corrections: March 1965; errata:

4th printing with corrections: December 1965; errata:

5th printing with corrections: August 1966

6th printing with corrections: November 1967

7th printing with corrections: May 1968

8th printing with corrections: 1969

9th printing with corrections: November 1970

10th printing with corrections: December 1972

Reprint edition by Dover Publications:

1st printing: 1965

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9th printing with additional corrections (based on 10th printing of NBS edition with corrections)

Related projects

Michael Danos and Johann Rafelski edited the Pocketbook of Mathematical Functions, published by Verlag Harri Deutsch in 1984. The book is an abridged version of Abramowitz's and Stegun's Handbook, retaining most of the formulas (except for the first and the two last original chapters, which were dropped), but reducing the numerical tables to a minimum, which, by this time, could be easily calculated with scientific pocket calculators. The references were removed as well. Most known errata were incorporated, the physical constants updated and the now-first chapter saw some slight enlargement compared to the former second chapter. The numbering of formulas was kept for easier cross-reference.

A digital successor to the Handbook, long under development at NIST, was released as the "Digital Library of Mathematical Functions" (DLMF) on 11 May 2010, along with a printed version, the NIST Handbook of Mathematical Functions, published by Cambridge University Press.

See also

Mathematical Tables Project, a 1938–1948 Works Progress Administration (WPA) project to calculate mathematical tables, including those later used in Abramowitz and Stegun's Handbook of Mathematical Functions

Bateman Manuscript Project

Gradshteyn and Ryzhik, a table of integrals.

Digital Library of Mathematical Functions (DLMF), from the National Institute of Standards and Technology (NIST), is intended to be a replacement for Abramowitz and Stegun's Handbook of Mathematical Functions

Prudnikov, Brychkov and Marichev (PBM)

Bronshtein and Semendyayev (BS)

Jahnke and Emde (JE)

Magnus and Oberhettinger (MO)

CRC Standard Mathematical Tables (CRC)

MAOL, a Finnish handbook for science

BINAS, a Dutch science handbook

Numerical analysis

Rubber book, a Handbook of Chemistry & Physics

Reference book

Handbook

Philip J. Davis, author of the Gamma function section and other sections of the book

Louis Melville Milne-Thomson, author of the book chapters on elliptic integrals and Jacobi elliptic functions

Boole's rule, a mathematical rule of integration sometimes known as Bode's rule, due to a typo in Abramowitz and Stegun (1972, p. 886) that was subsequently propagated.

On-Line Encyclopedia of Integer Sequences

Notes

References

Further reading

Abramowitz, Milton; Stegun, Irene Ann, eds. (1983) [June 1964]. Handbook of Mathematical Functions with Formulas, Graphs, and Mathematical Tables. Applied Mathematics Series. Vol. 55 (Ninth reprint with additional corrections of tenth original printing with corrections (December 1972); first ed.). Washington D.C.; New York: United States Department of Commerce, National Bureau of Standards; Dover Publications. ISBN 978-0-486-61272-0. LCCN 64-60036. MR 0167642. LCCN 65-12253.

Boisvert, Ronald F.; Lozier, Daniel W. (January 2001). "Handbook of Mathematical Functions" (PDF). In Lide, David R. (ed.). A Century of Excellence in Measurements Standards and Technology - A Chronicle of Selected NBS/NIST Publications 1901–2000. Washington, D.C., USA: U.S. Department of Commerce, National Institute of Standards and Technology (NIST) / CRC Press. pp. 135–139. ISBN 978-0-8493-1247-2. CODEN NSPUE2. NIST Special Publication 958. 20402–9325. Archived from the original (PDF) on 2011-10-14. Retrieved 2016-03-13. (NB. A history of the activities leading up to and surrounding the development of the Handbook.)

External links

A high quality scan of the book, in PDF and TIFF formats, hosted at the University of Birmingham, UK

The book in scanned format, now hosted at the University of British Columbia, CA (formerly at Simon Fraser University). # but this scan dropped tables of numbers. For example, the pages contained table of abscissas and weights of Gauss quadrature formulas are omitted. Only formulas are scanned.

Another scanned version by ConvertIt.com

numerical.recipes download - NBS, Tenth Printing, December 1972, with Corrections - PDF
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NIST Digital Library of Mathematical Functions, the digital successor of the Handbook